

SE1 Midterm Reviewer: Modules 1 and 2

CS0025 Software Engineering 1. Built from CS0024-M1S0-1, M1S1, M1S2, M1S3, CS0025-M2S0, M2S1, and M2S2. The blue boxes are memory aids I made; they are not from the slides. The red boxes flag places where the slides are easy to mix up.

What Is Inside

- **Part 1, Module 1:** Chapter 1 software and software engineering fundamentals; Chapter 2 systems and information systems; Chapter 3 the System Development Life Cycle.
- **Part 2, Module 2:** Chapter 1 project management, Gantt and PERT, and the project management phases; Chapter 2 analysis, system analysis, and computer science research trends.
- A one-page cram sheet at the end. Read it last, right before the exam.

Fast Memory Map

Topic	Hook
Software categories (7)	Sam Always Eats Eggs, Pizza, Waffles, Apples
SE layers (4)	Toddlers Make Pretty Quilts (Tools, Methods, Process, Quality Focus)
SE principles (7)	Real Kids Make Pretty Open Plans, Think!
IS types (7)	Record-keepers (TPS), Decision helpers (DSS, EIS, MIS), Coordinators (Workflow), Integrator (ERP), Advisor (Expert)
SDLC phases (7)	Please Always Design Decent, Test, Implement, Maintain
Gantt vs Fayol	Gantt = planning (chart); Fayol = five functions
Four P's	Plan, Process, People, Power
PM areas (10)	I Scoped This Cost Quickly, Procured Human Capital, Communicated Risks, Satisfied Stakeholders
PM phases (5)	Is Pizza Eaten Mid-Course?
Umbrella activities (8)	Tracking Risks Quietly, Technical Measures Control Reuse, Work
Analysis procedures (5)	Some People Really Love Deciding
CS research trends (5)	Amazing Big Computers Battle Cyberthreats

Part 1: Module 1, Software Engineering

Chapter 1: Software And Software Engineering

What Is A Software?

- It is a set of instructions or codes or computer program that when executed provides desired features, function and performance.
- They are data structures that enable the programs to adequately manipulate information (such as databases).
- It is a collection of computer programs, procedures and documentation that perform tasks on a computer system.

The Five Software Questions

Listed on the overview slide as **Software Guidelines**:

1. Why does it take long to finish a software?
2. Why are development costs so high?
3. Why can't we find all errors prior to release?
4. Why spend much time and effort to maintain an existing software?
5. Why do difficulties continue in measuring progress as software is developed and maintained?

Memory Aid

Tired Cats Eat Messy Meals = Time, Cost, Errors, Maintenance, Measuring progress.

All five questions are complaints about software: it is slow, costly, buggy, hard to maintain, and hard to measure.

The Seven Software Categories

Category	Definition
System Software	A collection of programs written to service other programs.
Application Software	Usually stand-alone programs that solve specific business needs.
Engineering/Scientific Software	A broad array of "number crunching" programs.
Embedded Software	Resides within a product or system, used to implement control features for the user and the system itself.
Product-line Software	Designed to provide specific capabilities for use by many customers.
Web/Mobile Application	Network-centric applications that encompass browser-based apps and/or mobile apps.
Artificial Intelligence Software	Makes use of non-numerical algorithms to solve complex problems not amenable to computation or forward analysis.

Memory Aid

Sam Always Eats Eggs, Pizza, Waffles, Apples = System, Application, Engineering/Scientific, Embedded, Product-line, Web/Mobile, AI.

Legacy Software

- These are older programs that were developed decades ago.
- They are continually modified today to meet changes in business requirements and computing platforms.
- They can be costly to maintain and risky to evolve.

"Infants" Software

Type	Definition
WebApps	Sophisticated computing tools that not only provide stand-alone function to users, but can also integrate databases and corporate applications.
Mobile Applications	Encompasses a user interface that takes advantage of the unique interaction mechanisms provided by the mobile platform.
Cloud Computing	Encompasses an infrastructure or "ecosystem" that enables any user from anywhere or anytime to use a computing device to share or use computing resources on a broad scale.
Product Line Software	A set of software-intensive systems that share a common, managed set of features satisfying the specific needs of a particular market segment.

Memory Aid

We Make Cool Products = WebApps, Mobile apps, Cloud computing, Product line software. These are called "infants" because they are the newer software categories, next to legacy software.

Software Realities

- Software has become deeply embedded in virtually every aspect of our lives.
- The IT requirements demanded by individuals or businesses grow in an increasingly complex terms.
- Individuals, businesses and even governments increasingly rely on software for processes and decision making.
- The perceived values of specific software application grows as its user base grows.

Memory Aid

Every Computer Relies on Value = Embedded everywhere, Complexity grows, Relies on it for decisions, Value grows with user base.

Software Engineering

Engineering is a discipline or profession of acquiring and applying technical, scientific or mathematical knowledge to design and implement structures, systems and processes to realize a desired objective. **Software Engineering** is a process or collection of methods and an array of tools that allow professionals to build high quality computer software.

It is a systematic, disciplined and quantifiable approach to development, operation and maintenance of software. Adaptability and agility are both required. The slide text says Software Engineering **is a layered technology and its foundation is the process layer**. It is a glue that holds the technology layers together and enables rational and timely development of computer software.

The Four Software Engineering Layers

The diagram, top to bottom: **Tools, Methods, Process, Quality Focus**.

Memory Aid

Toddlers Make Pretty Quilts = top to bottom, Tools, Methods, Process, Quality Focus. Quality Focus sits at the very bottom of the stack, so it is the **bedrock** everything else rests on.

Watch Out

The slide **text** (p12) says Software Engineering's **foundation is the process layer**. But the **diagram** (p13) stacks Quality Focus at the very bottom, under Process. If a question asks which layer is the **bedrock**, the diagram answer is **Quality Focus**; if it asks which layer is called the **foundation** in the slide text, the answer is **Process**. Both appear on the slides, so read the question's wording carefully.

The Software Process

Term	Definition
Process	A collection of activities, actions and tasks that are performed when some work product is to be created.
Activity	Strives to achieve a broad objective.
Action	A set of tasks that produce a major work product.
Task	Focuses on small, but well defined objectives to produce a tangible outcome.

Memory Aid

Please Arrange Action Tasks = Process, Activity, Action, Task, biggest to smallest.

The Process Framework (Five Framework Activities)

Activity	What It Means
Communication	Communicate and collaborate with the user to understand objectives.
Planning	Mapping the software project to create guidelines.

Modeling	A "sketch" of the project in order to understand the big picture. In other words, blueprints or software requirements design.
Construction	Building of the software project.
Deployment	Software is delivered to the customer who evaluates and provides feedback.

Memory Aid

Chatty People Make Cool Deliveries = Communication, Planning, Modeling, Construction, Deployment.

Software Engineering Practice (Four Practice Steps)

Step	What It Involves
Understand the Problem	Communication and Analysis (Research).
Plan a Solution	Modeling and Software Design (Designing).
Carry out the Plan	Code generation (Programming).
Examine the result for accuracy	Testing and Quality Assurance (Deliberate).

Memory Aid

Understand, Plan, Carry, Examine. This is the same shape as the scientific method: research, design, build, test.

The Seven Software Engineering Principles

Principle	What It Means
The Reason it all Exists	Provide value to its users.
KISS (Keep It Simple, STUPID!)	All design should be as simple as possible, but no simpler.
Maintain the Vision	It is essential for the success of the software project.
What you produce, others Consume	Always specify, design and implement knowing someone else will have to understand what you do.
Be open to the Future	Never design yourself in the corner.
Plan Ahead for Reuse	Reduces the cost and increases value of both reusable components and the system where it is incorporated.
Think!	Practice a complete and clear thought before taking action for better results.

Memory Aid

Real Kids Make Pretty Open Plans, Think! = Reason it Exists, KISS, Maintain vision, Produce for others to Consume, Open to the future, Plan ahead for reuse, Think.

Chapter 2: Systems And Information Systems

Systems

- It is a set of processes or integrated parts with entities each performing specific functions.
- It is a set of interacting or interdependent entities, real or abstract, forming an integrated whole.
- The concept of an "integrated whole" can also be stated in terms of a system embodying a set of relationships which are differentiated from relationships of the set to other elements, and from relationships between an element of the set and elements not a part of the relational regime.

System Types: Open Vs Closed

Type	Definition
Open Systems	Also called probabilistic systems. Outputs or results can't be determined precisely but can only be guessed. One can't predict such occurrences until the actual event arrives.
Closed Systems	Can be predicted with certainty. The measurement of the given input will actually let the output be predictable as it appears to be.

Watch Out

Open systems = **probabilistic**, their output can only be guessed, not calculated exactly.
Closed systems are the predictable ones. It is easy to mix these up because "open" sounds like it should be more knowable.

Information System

- It is a set of interrelated components that collect, store, process and help analyze, support decision-making and perform control functions in an organization.
- It handles the flow and maintenance of information that supports a business or some other operation.
- It is the system of people, data records and activities that process the data and information in a given organization, including manual processes or automated processes.

The IS Activities

Diagram: **Input** -> **Process** -> **Output**, with a **Feedback** loop back to Input.

Activity	What It Means
Input	Starts off with acquisition of raw data and is then introduced to the system.
Process	Involves data processing, classifying, and arranging, calculating data.
Output	The processed data becomes information.
Feedback	The evaluation.

Memory Aid

I Produce Output, Feedback = Input, Process, Output, Feedback, in that loop order.

IS Resource Approach

Diagram: **DATA** in the middle, surrounded by **Create** (data going in), **Destroy** (data going out), **Process**, **Retrieve**, and **Update**.

Memory Aid

DATA sits in the center. Around it are the classic **CRUD** actions (Create, Retrieve, Update, Destroy) plus **Process**, which appears twice on the diagram.

Information Management

- The system is managing information in its own way.
- Focusing on information, any kind of data presented in a form which is understandable or meaningful and helps make decisions.
- Today, Information Technology Enabled Services (ITES) consider information as a "utility", just like electricity or telephone, and the key factor is business strategy.

Six Good Information Characteristics

- Accuracy
- Timeliness
- Less Uncertainty
- An element of surprise value
- Should aid in decision-making
- Should update knowledge

Memory Aid

Any Timely Lead Adds Decisions, Updates = Accuracy, Timeliness, Less uncertainty, Element of surprise, Aid decision-making, Update knowledge.

IS Business Perspective

Diagram: a circle with **Information System** in the centre, surrounded by **Organizational structure and system, Environmental Factors, Management Policy, and Technology Development**.

Memory Aid

Our Era Means Tech = Organizational structure, Environmental factors, Management policy, Technology development, all circling the IS.

IS Key Components

Diagram: **DATA, PEOPLE, TECHNOLOGY, EQUIPMENT, PROGRAMS, PROCEDURES.**

Memory Aid

Data People Trust Equipment, Programs, Procedures = Data, People, Technology, Equipment, Programs, Procedures.

IS Major Components

Component	Definition
Data	Consists of raw facts and is an important component of an IS. It has two sources: external and internal.
Database	Is a collection of all relevant and related organized data in an integrated file.
Process	An important component of IS that generates the most useful information for decision-making.
Information	Consists of analyzed facts that was processed from a data source and is an output of an IS.

Memory Aid

Data Drives Processed Information = Data, Database, Process, Information.

The Seven Types Of Information Systems

Type	Who Uses It / Purpose	Example
Transaction Processing System (TPS)	Used primarily for record keeping required in any organization to conduct business. Used for periodic report generation in a scheduled manner, and for reports on demand as well as exception reports.	Sales order entry, payroll, shipping records.
Decision Support System (DSS)	Serves the management of an organization with sophisticated data analysis tools that support and assist all aspects of problem-specific decision-making. Used when the problem is complex and information is difficult to obtain and use.	Developed with decision-makers; may use external data such as current stock prices.
Executive Information System (EIS)	Used by senior managers, so it must be easy to use without assistance. Can do trend analysis, exception reporting, and drill-down. Has online analysis tools accessing a broad range of internal and external data.	Results shown in graphical form tailored to the executive's needs.
Management Information System (MIS)	Provides management routine summary of basic operations; consolidates data on sales, production, etc. Provides routine information to managers and decision	May support marketing, production, finance.

	makers to increase operational efficiency.	
Workflow System	A rule-based management system that directs, coordinates and monitors the execution of an interrelated set of tasks arranged to form a business process. May be Internet-based, combined with e-mail, or based on server architecture using a database or file server.	See the three workflow types below.
Enterprise Resource Planning (ERP)	A business process management software that allows an organization to use a system of integrated programs capable of managing a company's vital business operations.	For an entire multi-site, global organization.
Expert Systems	Has the ability to make suggestions and act like an expert in a particular field of an organization.	Has an extensive knowledge base.

Memory Aid

Group the seven by job: **record-keepers** (TPS), **decision helpers** (DSS, EIS, MIS), **coordinator** (Workflow), **integrator** (ERP), and **advisor** (Expert Systems, which make suggestions from a knowledge base).

Watch Out

DSS helps decide, but does not make the decision itself. It is developed with the decision-makers and assists the decision-making process only.

Watch Out

EIS is also called the Executive Support System. If a question gives that name instead of the acronym, it is still the EIS.

Three Types Of Workflow Systems

- **Administrative Workflow Systems** - focus on the tracking of expense reports, travel requests, messages.
- **Ad-hoc Workflow System** - deals with the shaping of product, sales proposals and strategic plans.
- **Production Workflow Systems** - are concerned with mortgage loans and insurance claims.

Chapter 3: System Development Life Cycle (SDLC)

What Is SDLC?

- Also referred to as the application development life-cycle, is a process for planning, creating, testing, and deploying an information system.
- The systems development life cycle concept applies to a range of hardware and software configurations, as a system can be composed of hardware only, software only, or a combination of both.
- SDLC is used during the development of an IT project and describes the different stages involved in the project from the drawing board, through the completion of the project.
- SDLC describes the stages involved in an information system development project, from an initial feasibility study through maintenance of the completed application. It can apply to technical and non-technical systems involving hardware and software.
- In SDLC, documentation is crucial, regardless of the type of model chosen for any application, and is usually done in parallel with the development process.

The Seven SDLC Phases

Cycle on the slide: **Planning -> Analysis -> Design -> Development -> Testing -> Implementation -> Maintenance -> back to Planning.**

Phase	What Happens	Key Detail
1. Planning	Identifies whether or not there is a need for a new system. A preliminary plan (feasibility study) for a company's business initiative to acquire resources to build, modify or improve a service.	Finds the scope of the problem and determines solutions considering resources, costs, time, benefits.
2. System Analysis	The business works on the source of the problem or the need for change. Possible solutions are submitted and analyzed to identify the best fit for the project's goals. Teams consider the functional requirements of the project.	Tools: CASE (Computer Aided Systems/Software Engineering), requirements gathering, structured analysis.
3. System Design	Describes in detail the necessary specifications, features and operations that will satisfy the functional requirements of the proposed system. End users discuss and determine their specific business information needs.	Considers essential components (hardware and/or software), networking capabilities, processing, and procedures.
4. System Development	The real work begins: a programmer, network engineer and/or database developer are brought on to do the major work on the project.	Uses a flow chart to organize the process; signifies the start of production; training can be a big benefit here.
5. System Testing	Systems integration and system testing of programs and procedures, normally	May be repeated to check for errors, bugs and

	carried out by a QA professional to determine if the design meets the initial goals.	interoperability; includes verification and validation.
6. System Implementation	The slide says this is when the majority of the code for the program is written. Also involves the actual installation of the newly developed system, moving data and components from the old system via a direct cutover .	Cutover typically happens during off-peak hours to minimize risk.
7. System Maintenance	Involves maintenance, regular operations and required updates.	End users can fine-tune the system; new updates usually require looking back at the first phase, beginning the cycle again.

Memory Aid

Please Always Design Decent, Test, Implement, Maintain = Planning, Analysis, Design, Development, Testing, Implementation, Maintenance.

Watch Out

Which phase writes the code? On these slides (and their source, Gillis and Rouse, TechTarget), **System Implementation** is "when the majority of the code for the program is written" and is also when the system is installed by **direct cutover** during off-peak hours. **System Development** is where programmers, network engineers and database developers are **brought on**, a flow chart organizes the process, and production starts. Many other textbooks call coding "development", so if a choice says "majority of the code is written", answer **Implementation** for this course.

System Development Life Cycle Vs Software Development Life Cycle

System Development Life Cycle	Software Development Life Cycle
Involves end-to-end People, process, Software/Technology deployment. This includes Change Management, training, Organizational updates.	Only looks at software components development planning, technical architecture, software quality testing and deployment of working software.
It is a much broader term which is a superset to the above and is a larger part of the development phase which includes many other approaches and end phases.	The meaning of the process gets limited to a certain boundary beyond which it ceases to provide relevance.
It is about implementing hardware and software in a phased manner systematically. Here both the hardware and the software are considered as a system.	It is about building a software ("only") in a phased approach systematically. Hence, it's just part of the other SDLC, in particular during the Development Phase.

Watch Out

System SDLC is the superset. It covers people, process, and technology end to end (hardware AND software, as a system). **Software SDLC is the narrower one:** it is only part of the Development Phase of the System SDLC, and only covers building the software itself.

Part 2: Module 2, Project Management

Chapter 1: Project Management

What Is A Project?

A project is a **temporary endeavor** designed to produce a unique product or service with a defined beginning and end (usually time-constrained, and often constrained by funding or staffing) undertaken to meet unique goals and objectives, to bring about beneficial change or added value. It is an activity to meet the creation of a unique product or service, and thus activities that are undertaken to accomplish **routine activities cannot be considered projects**.

Project Application

- Schools and Universities
- Civil and Industry (Private)
- Military and State (Government)
- Computer Software (IT-based)

Memory Aid

Schools, Construction, Military, Computers = Schools/Universities, Civil/Industry (Private), Military/State (Government), Computer Software (IT-based).

Project Management

Project management is the **practice** of initiating, planning, executing, controlling, and closing the work of a team to achieve specific goals and meet specific success criteria at the specified time. The primary challenge is to achieve all of the project goals within the given constraints. This is usually described in **project documentation**, created at the beginning of the development process.

Project Manager

A professional in the field of project management, in-charge of people and responsibilities such as planning, execution, controlling, and closing of any project. A project manager needs to understand the order of execution of a project to schedule it correctly, as well as the time necessary to accomplish each individual task. They are the people **accountable for accomplishing the stated project objectives**.

Quote from the slide: "**Project Management is not a tool or a person, it's a practice.**"

Origins Of Project Management

As a discipline, project management developed from several fields of application including civil construction, engineering, and heavy defense activity.

Forefather / Era	Contribution
Henry Gantt	Called the father of planning and control techniques , famous for his

	use of the Gantt Chart .
Henri Fayol	Creator of the five management functions that form the foundation of the body of knowledge associated with project and program management.
The 1950s	Marked the beginning of the modern project management era, where core engineering fields came together to work as one. Project management became recognized as a distinct discipline arising from the management discipline with engineering mode.
CPM and PERT	Two mathematical project-scheduling models developed in that era: the Critical Path Method (CPM) and the Program/Project Evaluation and Review Technique (PERT) .

Watch Out

Gantt = father of planning (and control techniques), remembered for the Gantt Chart.
Fayol = the five management functions. They are the two separate forefathers; do not swap their contributions.

Gantt Chart Vs PERT Chart

	Gantt Chart	PERT Chart
What it is	A bar chart of the schedule: each task is a bar along a timeline (the slide's example was made in Microsoft Project).	A project management tool used to schedule, organize, and coordinate tasks within a project.
What it shows	Red marks indicate the critical path , or the longest stretch of the project. Columns: ID, Task Name, Predecessors, Duration.	A network diagram of tasks and their dependencies, used to analyze the tasks involved in completing a project.
Main purpose	Visualize schedule and the critical path.	Identify the minimum time needed to complete the total project, especially the time needed to complete each task.

The PERT Example From The Slide

Nodes: **10, 20, 30, 40, 50**. Activities:

- A: 10 -> 30, t = 3 months
- B: 10 -> 20, t = 4 months
- C: 20 -> 50, t = 3 months
- D: 30 -> 40, t = 1 month
- E: 30 -> 50, t = 3 months
- F: 40 -> 50, t = 3 months

The longest path is **10-30-40-50 = 7 months**, which ties with **10-20-50 = 7 months**. Both paths take 7 months, so either can be the critical path in this example.

Project Management Application

Project management methods can be applied to any project. It is often customized to a specific type of project based on size, nature and industry. For example, the construction industry, which focuses on the delivery of things like buildings, roads, and bridges, has developed its own specialized form of project management that it refers to as **construction project management**, in which project managers can become trained and/or certified.

The Four P's Approach

P	Meaning
Plan	The planning and forecasting activities.
Process	The overall approach to all activities and project governance.
People	Including dynamics of how they collaborate and communicate.
Power	Lines of authority, decision-makers, organograms, policies for implementation and the like.

Memory Aid

The name is already the mnemonic: **Plan, Process, People, Power**, all four P's.

The Ten Project Management Areas

- Integration
- Scope
- Time
- Cost
- Quality
- Procurement
- Human resources
- Communications
- Risk management
- Stakeholder management

Memory Aid

I Scoped This Cost Quickly, Procured Human Capital, Communicated Risks, Satisfied Stakeholders = Integration, Scope, Time, Cost, Quality, Procurement, Human resources, Communications, Risk management, Stakeholder management.

The Five Project Management Phases

Flow on the slide: **Initiation -> Planning and Design -> Executing <-> Monitoring and Controlling -> Closing.**

Memory Aid

Is Pizza Eaten Mid-Course? = Initiation, Planning and design, Execution, Monitoring and controlling, Closing. Execution and Monitoring/Controlling loop back and forth ("mid-course") before Closing.

Phase	Description	Process-Group Items From The Diagram
1. Initiation	Determines the nature and scope of the project. If this stage is not performed well, it is unlikely the project will succeed in meeting the business' needs. Needs an understanding of the business environment and all necessary controls incorporated.	Initiating Process Group -> Scope: Develop Charter; Scope: Develop Preliminary Scope Statement.
2. Planning and Design	After initialization, the project is planned to an appropriate level of detail. The main purpose is to plan time, cost and resources adequately, to estimate the work needed, and to effectively manage risk during execution.	(No diagram items given beyond the phase flow.)
3. Execution	One must know what are the planned terms to be executed. Ensures the project management plan's deliverables are executed accordingly, with proper allocation, co-ordination and management of human and other resources. Output: the project deliverables.	Planning Process Group -> Integration: Project Management Plan Execution; Quality: Quality Assurance; Communication: Information Distribution; Communication: Team Development; Procurement: Plan Contracting -> Source Selection -> Contract Administration. Loops with the Controlling Process Group.
4. Monitoring and Controlling	Processes performed to observe project execution so potential problems can be identified in a timely manner and corrective action taken when necessary. Key benefit: performance is observed and measured regularly to identify variances from the plan.	Communications: Performance Reporting; Integration: Integrated Change Control; Scope: Scope Verification; Scope: Scope Change Control; Time: Schedule Control; Cost: Cost Control; Quality: Quality Control; Risk: Risk Monitoring and Control (the slide's diagram labels these last two boxes "Procurement"; they belong to Quality and Risk). Feeds the Closeout Process Group.
5. Closing	Includes the formal acceptance and ending of the project. Administrative	Controlling Process Group -> Procurement: Contract Closure;

	activities include archiving files and documentation.	Communication: Close Project. Consists of Contract closure (settle and close each contract) and Project close (finalize all activities to formally close the project or phase).
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Project Documentation

Documenting everything within a project is key to being successful. To maintain budget, scope, effectiveness and pace, a project must have physical documents pertaining to each specific task. With correct documentation, it is easy to see whether or not a project's requirement has been met, and it provides information regarding what has already been completed.

Documentation throughout a project provides a documented trail for anyone who needs to go back and reference past work. In most cases, documentation is the most successful way to monitor and control the specific phases of a project. If performed correctly, it can be the **backbone to a project's success**.

Five Project Characteristics

- Projects should always have a specific start and end dates.
- They are performed and completed by a group of people working as one.
- The output should deliver a working/unique product or service.
- They are temporary in nature.
- They are progressively elaborated.

Memory Aid

Smart Groups Usually Try Everything = Start/end dates, Group of people, Unique output, Temporary, Elaborated progressively.

The Eight Umbrella Activities

Activity	What It Means
Project Tracking and Control	Assess the progress of the project plan.
Risk Management	Assess risks that may affect the product outcome.
Quality Assurance	Conduct activities to ensure product quality.
Technical Review	Assess the work product to identify errors beforehand.
Measurement	Defines and collects process to deliver and meet the needs of the stakeholders.
Configuration Management	Manages the effects of changes in the software process.
Reusability Management	Defines criteria to establish mechanisms for reusable components.
Work Product Preparation and Production	Encompasses activities required to create work products such as models or documentation.

Memory Aid

Tracking Risks Quietly, Technical Measures Control Reuse, Work = Tracking and Control, Risk management, Quality assurance, Technical review, Measurement, Configuration management, Reusability management, Work product preparation.

Chapter 2: Analysis And Computer Science Research Trends

Analysis Vs Synthesis

Analysis is defined as "the procedure by which we break down an intellectual or substantial whole into parts," while **synthesis** means "the procedure by which we combine separate elements or components in order to form a coherent whole." System analysis researchers apply methodology to the systems involved, forming an overall picture.

System Analysis

It is the process of studying a procedure or business in order to identify its goals and purposes and create systems and procedures that will achieve them in an efficient way. It is also a problem-solving technique that breaks down a system into its component pieces for the purpose of studying how well those component parts work and interact to accomplish their purpose.

The Analysis Phase

The analysis phase involves gathering requirements for the system. At this stage, business needs are studied with the intention of making business processes more efficient. The system analysis phase focuses on what the system will do, in an effort that views all stakeholders as viable sources of information. A significant amount of time is spent talking with stakeholders and reviewing their input.

The Five Analysis Phase Procedures

Procedure	What It Means
1. Scope Definition	Clearly defined objectives and requirements necessary to meet a project's requirements as defined by its stakeholders.
2. Problem Analysis	The process of understanding problems and needs and arriving at solutions that meet them.
3. Requirements Analysis	Determining the conditions that need to be met.
4. Logical Design	Looking at the logical relationship among the objects.
5. Decision Analysis	Making a final decision.

Memory Aid

Some People Really Love Deciding = Scope definition, Problem analysis, Requirements analysis, Logical design, Decision analysis.

Computer Science Research Trends

Computer science graduates have some of the highest starting salaries out there and are in such high demand that they can afford to be picky about the type of job and industry they opt for. If you are interested in pursuing a career in computer science, it's important to stay up to date with the latest trends in computer science research, to make an informed choice about where to head next.

The Five CS Research Trends (2019 to 2020)

Trend	Description
Artificial Intelligence and Robotics	One of the most controversial and intriguing areas of CS research. Still in its early stages, but tech giants like Facebook, Google and IBM are investing huge amounts of money and resources into AI research. No shortage of opportunities for real-world applications and breakthroughs.
Big Data Analytics	A surge in demand for experts in this field, with doubled efforts by brands and agencies to boost salaries and attract data science talents. From banking to healthcare, big data analytics is everywhere as companies try to make better use of enormous datasets.
Computer-assisted Education	The use of computers and software to assist education and/or training. Brings many benefits, such as personalized instruction for students with learning disabilities, letting them learn at their own pace. Still growing but promising, allowing active, independent, play-based learning.
Bioinformatics	An application of big data, or the use of programming and software development to build enormous datasets of biological information, carrying enormous potential. Linking big pharmaceutical companies with software companies, it offers good job prospects for CS researchers interested in biology, medical technology, and pharmaceuticals.
Cyber Security	We live in a hyper-connected world where everything, from banking to dating to governmental infrastructure, is done online. Data protection is no longer optional, for either individuals or nations, making this a growing strand of CS research.

Memory Aid

Amazing Big Computers Battle Cyberthreats = AI and robotics, Big data analytics, Computer-assisted education, Bioinformatics, Cyber security.

Top Research Titles By Trend (2019, Low Detail)

The slides list these 2019 research titles under each trend.

Research Area	Titles Listed
AI and Robotics	The Lottery Ticket Hypothesis: Finding Sparse, Trainable Neural Networks; Challenging Common Assumptions in the Unsupervised Learning of Disentangled Representations; Meta-Learning Update Rules for Unsupervised Representation Learning.
Big Data Analytics	An integrated parallel big data decision support tool using the W-CLUS-MCDA: A multi-scenario personnel assessment; Effect of E-customization Capability on Financial Performance of Commercial Banks in Kenya; Interaction of East Bay Area Voters with California

	Death Penalty Ballot Initiatives: a Study; Command Decision: Ethical Leadership In The Information Environment.
Computer-Assisted Education	Qualitative case studies of innovative pedagogical practices using ICT; What strategies are effective for formative assessment in an e-learning environment?; Web-based Assessment and Test Analyses (WATA) system: development and evaluation; Computer-Assisted Learning in Orthodontic Education: A Systematic Review and Meta-Analysis.
Bioinformatics	A Survey for Escherichia coli Virulence Factors in Asymptomatic Free-Ranging Parrots; Immunoreactivity of the 14F7 Mab Raised against N-Glycolyl GM3 Ganglioside in Epithelial Malignant Tumors from Digestive System; Investigating the effects of external fields polarization on the coupling of pure magnetic waves in the human body in very low frequencies.
Cyber Security	Intelligence Gathering And Social Media-Based Sentiment Correlation; High Accuracy Phishing Detection Based on Convolutional Neural Networks; Understanding Security Policies in the Cyber Warfare Domain Through System Dynamics.

One-Page Cram Sheet

Topic	Remember
5 software questions	Time, Cost, Errors before release, Maintenance, Measuring progress
7 software categories	System, Application, Engineering/Scientific, Embedded, Product-line, Web/Mobile, AI
SE layers, top to bottom	Tools, Methods, Process, Quality Focus (Quality Focus = bedrock, Process = text's "foundation")
Process framework (5)	Communication, Planning, Modeling, Construction, Deployment
7 SE principles	Reason it exists, KISS, Maintain vision, Produce for others, Open to future, Plan for reuse, Think
Open vs Closed systems	Open = probabilistic (guess only); Closed = predictable
IS activities	Input, Process, Output, Feedback
6 good info traits	Accuracy, Timeliness, Less uncertainty, Surprise value, Aid decisions, Update knowledge
7 IS types	TPS, DSS, EIS (= Executive Support System), MIS, Workflow, ERP, Expert Systems
DSS trap	DSS helps decide, does NOT make the decision
7 SDLC phases	Planning, Analysis, Design, Development, Testing, Implementation, Maintenance
Implementation trap	This course: majority of code written + direct cutover = Implementation; Development = team brought on, flow chart, start of production
System vs Software SDLC	System SDLC = superset, people+process+tech; Software SDLC = only software, part of the Development Phase
Gantt vs Fayol	Gantt = father of planning and control (Gantt Chart); Fayol = five management functions
PERT example	Longest path 10-30-40-50 = 7 months, ties with 10-20-50 = 7 months
Four P's	Plan, Process, People, Power
10 PM areas	Integration, Scope, Time, Cost, Quality, Procurement, HR, Communications, Risk, Stakeholder
5 PM phases	Initiation, Planning and Design, Execution, Monitoring and Controlling, Closing
5 project characteristics	Start/end dates, group effort, unique output, temporary, progressively elaborated
8 umbrella activities	Tracking and Control, Risk Mgmt, QA, Technical Review, Measurement, Configuration Mgmt, Reusability Mgmt, Work Product Preparation
Analysis vs Synthesis	Analysis breaks a whole into parts; Synthesis combines parts into a whole
5 analysis procedures	Scope Definition, Problem analysis, Requirements analysis, Logical design, Decision analysis
5 CS research trends	AI and robotics, Big data analytics, Computer-assisted education, Bioinformatics, Cyber security

